Research Proposal

Department of Computer Science and Engineering

Topic: Adaptive Bitrate Streaming for Cloud Gaming

Focus: Proactive Optimization over Unstable 5G Networks

1. Research Title

"Proactive Bitrate Adaptation using Deep Reinforcement Learning for Resilient Cloud Gaming over Unstable 5G Networks"

2. Problem Statement

The deployment of Cloud Gaming on 5G networks faces three critical bottlenecks:

- **Reactive Latency:** Conventional Adaptive Bitrate (ABR) algorithms (e.g., BOLA, Youtube Blur) are reactive. They adjust quality *after* network congestion is detected, leading to noticeable "stutter" during gameplay.
- **5G mmWave Sensitivity:** 5G signals, particularly in high-frequency bands, are highly susceptible to physical obstructions and "handover" drops, causing sudden throughput fluctuations.
- **The Gaming Constraint:** Unlike standard video streaming (Netflix/YouTube), cloud gaming cannot use large buffers to hide network instability. Every millisecond of delay (Interaction Latency) directly impacts the user's ability to play.

3. Objectives

- **Predictive Modeling:** To develop an AI-based system that predicts 5G signal fading 100–300ms in advance using real-time network telemetry.
- **QoE Optimization:** To implement a specialized Quality of Experience (QoE) metric that prioritizes **Motion-to-Photon (MTP)** latency over visual resolution during high-motion scenes.
- **Seamless Handover Management:** To minimize frame-dropping and "freezing" during 5G cell tower transitions in high-mobility scenarios (e.g., gaming while traveling).

4. Proposed Methodology & Selected Model

Selected Model: Asynchronous Advantage Actor-Critic (A3C) with LSTM Layers.

5. Dataset & Tools

- **5G Traces:** The **LUMOS 5G Dataset**, which contains real-world 5G throughput and signal strength data from various mobility patterns (walking, driving, stationary).
- **Game Engine Traces:** Video complexity traces from open-source gaming platforms like **GamingAnywhere**.
- **Simulation Environment: NS-3 (5G-LENA)** for network simulation and **PyTorch** for training the DRL agent.

6. Evaluation Metrics

To validate the research, the following metrics will be used:

1. **MTP Latency (ms):** The primary KPI—measuring the time from user input to screen update.
2. **VMAF (Video Multi-Method Assessment Fusion):** To ensure the visual quality remains acceptable during bitrate downscaling.
3. **Stall Ratio:** The percentage of time the game stream is frozen due to network drops.
4. **QoE Score:** A weighted sum of quality, smoothness, and latency penalties.